Elasticity

Chapter 19

Price Elasticity of Demand

- A measure of the responsiveness of quantity demanded to changes in price.
- Measured by dividing the percentage change in the quantity demanded of a good by the percentage change in its price.
- Economists compute price elasticity of demand using midpoints as the base values of changes in prices and quantities demanded.

Computing Elasticity of Demand

We divide the change in quantity demanded by the average quantity demanded, all of which is then divided by the change in price divided by the average price.

$$E_d = \frac{\text{Percentage change in quantity demand}}{\text{Percentage change in price}} = \frac{\%\Delta Q_d}{\%\Delta P}$$

$$E_d = \frac{\dfrac{\Delta Q_d}{Q_{d\ \text{average}}}}{\dfrac{\Delta P}{P_{\text{average}}}}$$

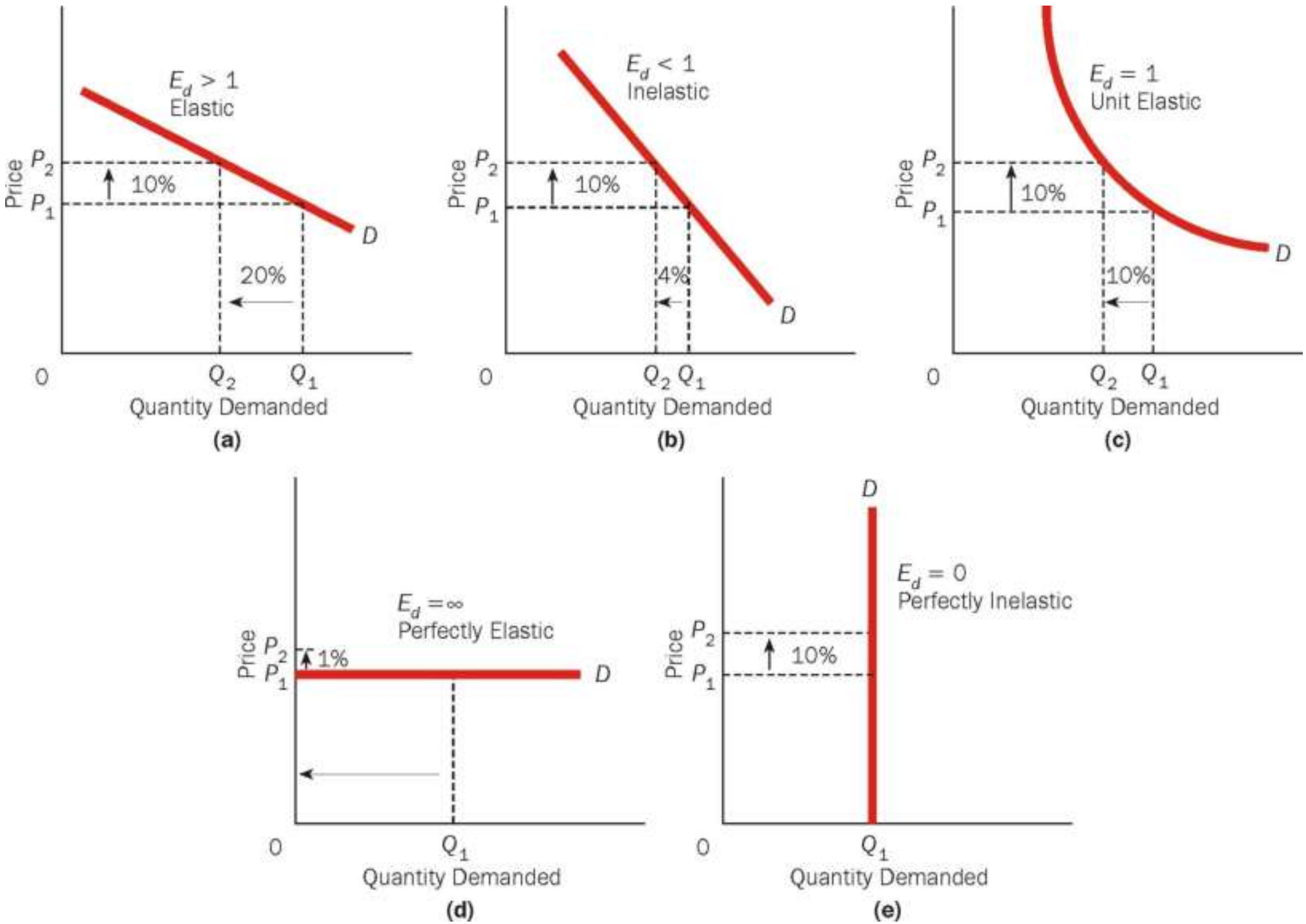
Perfectly Elastic to Perfectly Inelastic Demand

- Elastic Demand ($E_d > 1$): the numerator is greater than the denominator, the coefficient is greater than 1 and demand is elastic.
- Inelastic Demand ($E_d < 1$): the numerator is less than the denominator, the coefficient is less than 1, and demand is inelastic.
- Unit Elastic Demand (E_d=1): If the numerator and denominator are the same, the coefficient is equal to one. The quantity demanded changes proportionally to a change in price.
- Perfectly Elastic Demand (Ed = ∞) If the quantity demanded is extremely responsive to a change in price.
- Perfectly Inelastic Demand (Ed = 0) If the quantity demanded is completely unresponsive to changes in price, demand is perfectly inelastic. A change in price causes no change in quantity demanded.

Perfectly Elastic and Perfectly Inelastic Demand

Elasticity Coefficient	Responsiveness of Quantity Demanded to a Change in Price	Terminology
$E_d > 1$	Quantity demanded changes proportionately more than price changes: $\%\Delta Q_d > \%\Delta P$.	Elastic
$E_d < 1$	Quantity demanded changes proportionately less than price changes: $\%\Delta Q_d < \%\Delta P$.	Inelastic
$E_d = 1$	Quantity demanded changes proportionately to price change: $\%\Delta Q_d = \%\Delta P$.	Unit elastic
$E_d = \infty$	Quantity demanded is extremely responsive to even very small changes in price.	Perfectly elastic
$E_d = 0$	Quantity demanded does not change as price changes.	Perfectly inelastic

Price Elasticity of Demand

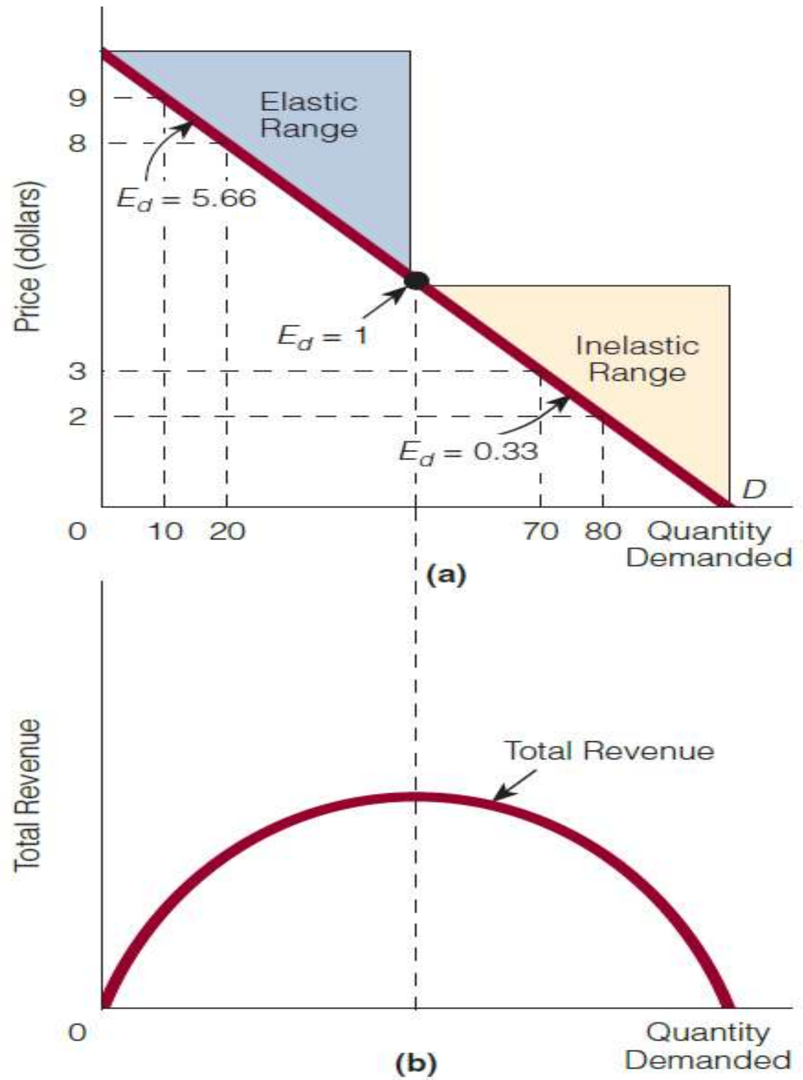


Slope Vs Elasticity

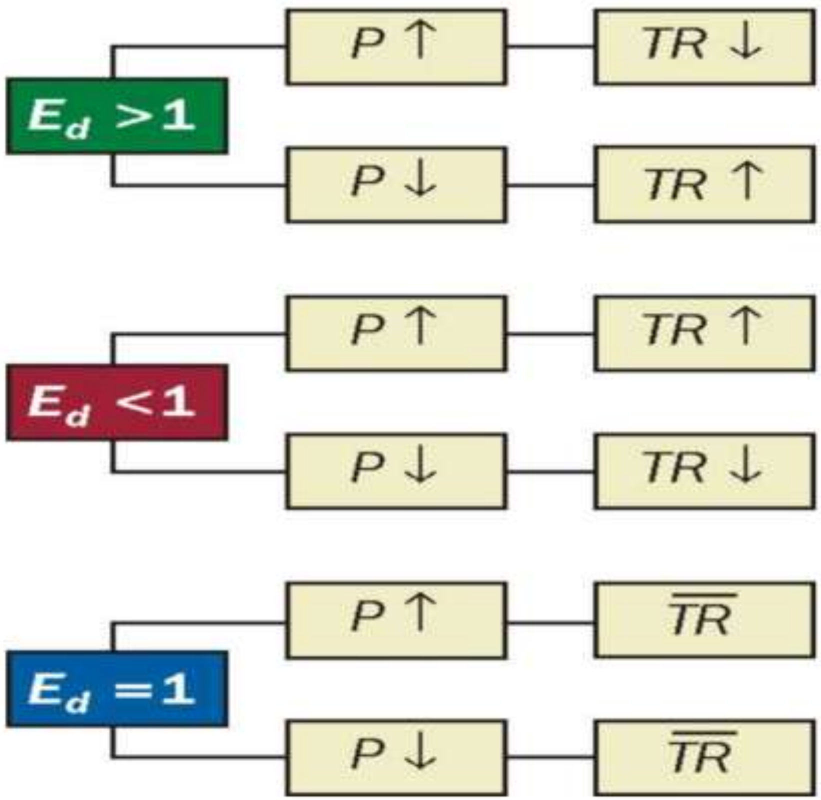
$$\text{Slope} = \frac{\Delta\text{Variable on vertical axis}}{\Delta\text{Variable on horizontal axis}}$$

$$E_d = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

Price Elasticity of Demand and Total Revenue



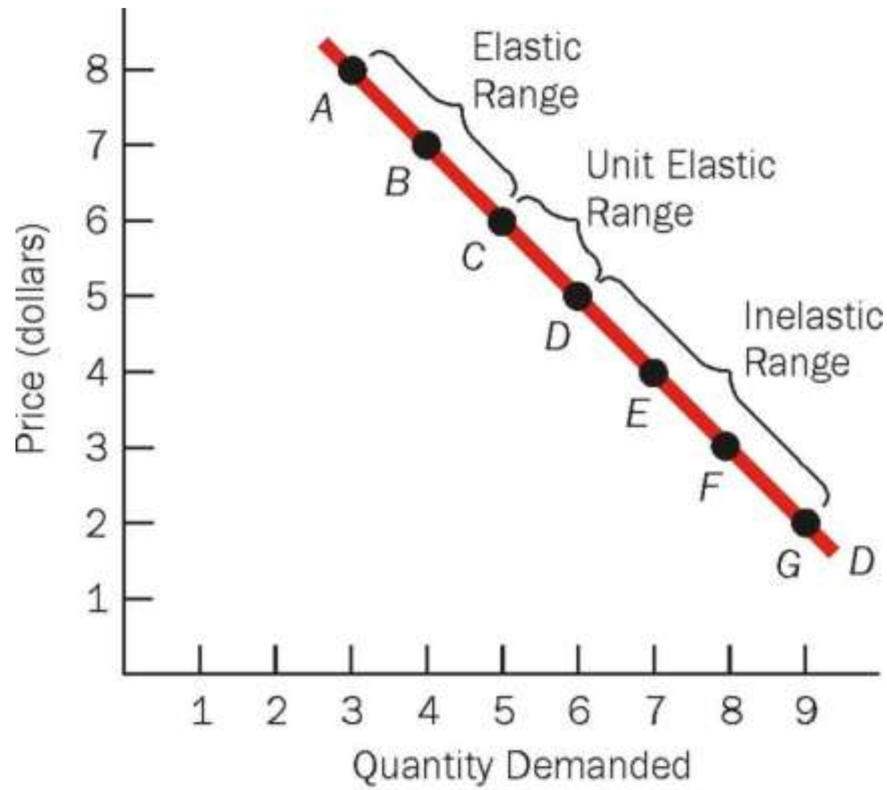
Price Elasticity of Demand and Total Revenue



Price Elasticity of Demand Along a Demand Curve

(1) Point	(2) Price	(3) Quantity Demanded	(4) E_d
A	$8	3	
			2.14
B	7	4	
			1.44
C	6	5	
			1.00
D	5	6	
			0.69
E	4	7	
			0.47
F	3	8	
			0.29
G	2	9	

(a)



(b)

Determinants of Price Elasticity on Demand

- **Number of Substitutes**: The more substitutes for a good, the higher the price elasticity of demand; the fewer substitutes for a good, the lower the price elasticity of demand.

The more broadly defined the good, the fewer the substitutes; the more narrowly defined the good, the greater the substitutes.

- **Necessities vs Luxuries**
- **Percentage of One's Budget Spent on the Good**: The greater the percentage of one's budget that goes to purchase a good, the higher the price elasticity of demand; the smaller the percentage of one's budget that goes to purchase a good, the lower the elasticity of demand.
- **Time**: The more time that passes, the higher the price elasticity of demand for the good; the less time that passes, the lower the price elasticity of demand for the good.

Cross Elasticity of Demand

Measures the responsiveness in the quantity demanded of one good to changes in the price of another good. Defined as the percentage change in the quantity demanded of one good divided by the percentage change in the price of another good.

$$E_c = \frac{\text{Percentage change in quantity demanded of one good}}{\text{Percentage change in price of another good}}$$

This concept is often used to determine whether two goods are substitutes or complements and the degree to which one good is a complement to or substitute for another.

$$E_c > 0 \rightarrow \text{Goods are substitutes}$$

$$E_c < 0 \rightarrow \text{Goods are complements}$$

Income Elasticity of Demand

Measures the responsiveness of quantity demanded to changes in income. Define as the percentage change in quantity demanded of a good divided by the percentage change in income.

$$E_y > 0 \rightarrow \text{Normal good}$$

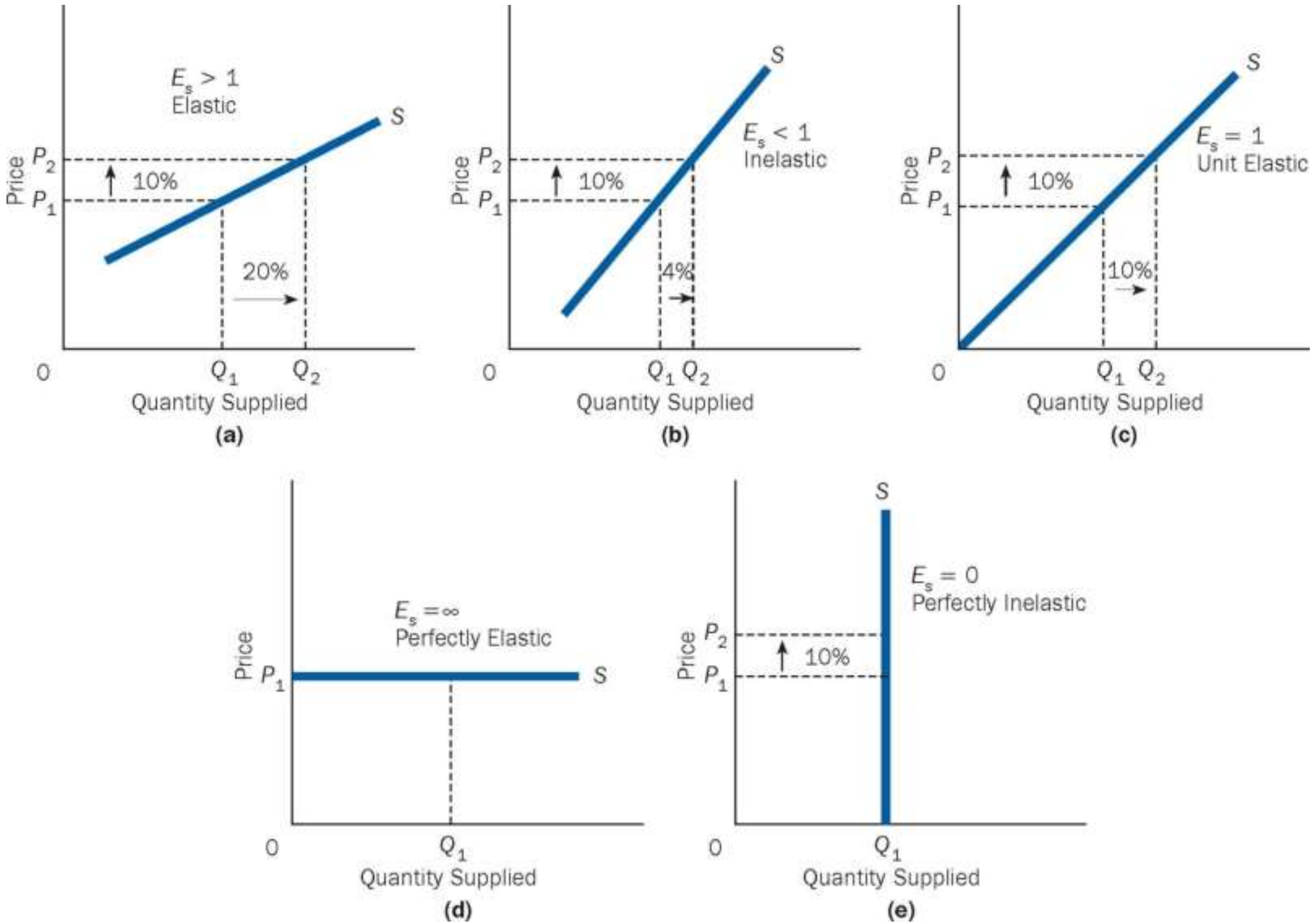
$$E_y < 0 \rightarrow \text{Inferior good}$$

- If $E_y > 1$, demand is considered to be income elastic.
- If $E_y < 1$, demand is considered to be income inelastic.
- If $E_y = 1$, demand is considered to be unit elastic

Price Elasticity of Supply

- Measures the responsiveness of quantity supplied to changes in price.
- Defined as the percentage change in quantity supplied of a good divided by the percentage change in the price of the good.
- Supply can be classified as elastic, inelastic, unit elastic, perfectly elastic, or perfectly inelastic.

Price Elasticity of Supply



Price Elasticity of Supply and Time

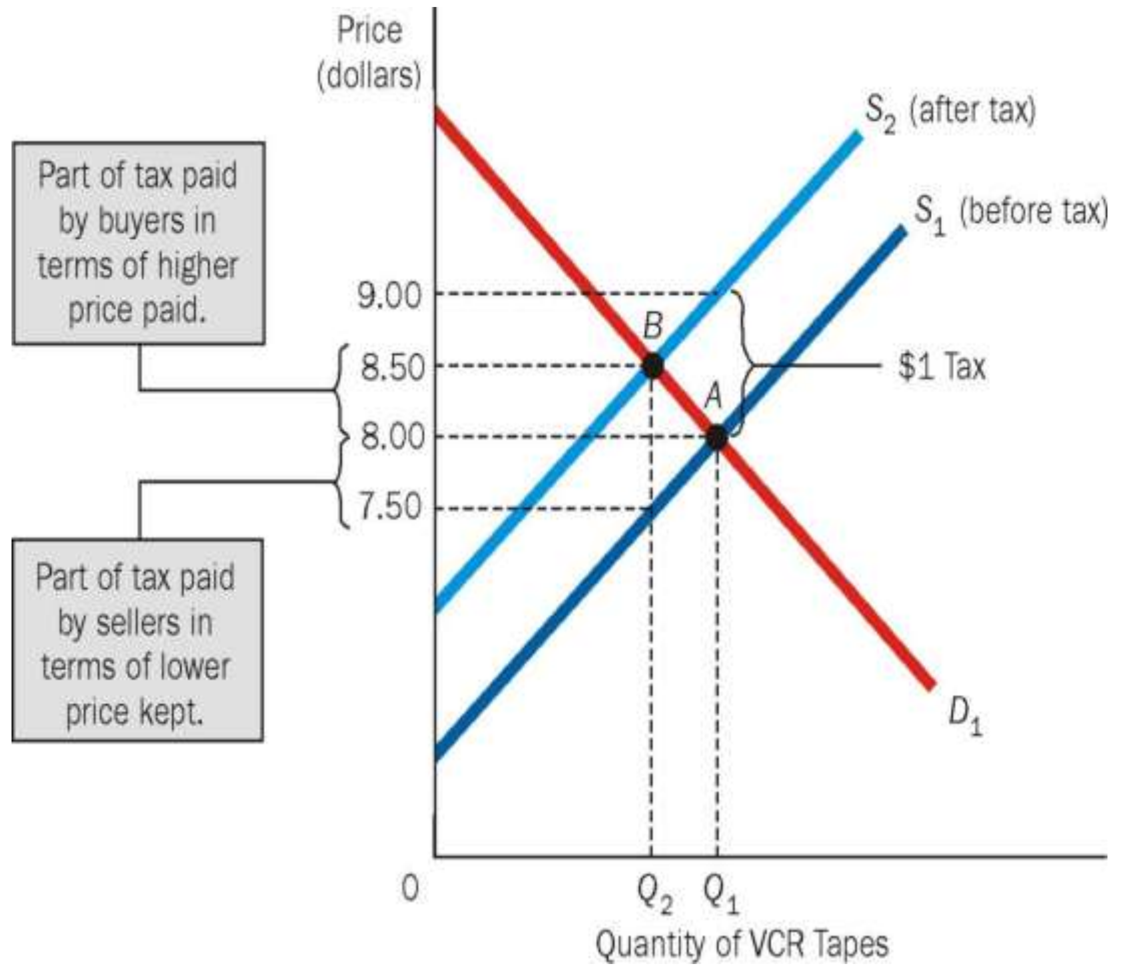
- The longer the period of adjustment to a change in price, the higher the price elasticity of supply.
- Additional production takes time.
- Reducing production takes time.

Summary of the Four Elasticity Concepts

Type	Definition	Possibilities	Terminology
Price elasticity of demand	$\frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$	$E_d > 1$ $E_d < 1$ $E_d = 1$ $E_d = \infty$ $E_d = 0$	Elastic Inelastic Unit elastic Perfectly elastic Perfectly inelastic
Cross elasticity of demand	$\frac{\text{Percentage change in quantity demanded of one good}}{\text{Percentage change in price of another good}}$	$E_c < 0$ $E_c > 0$	Complements Substitutes
Income elasticity of demand	$\frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$	$E_y > 0$ $E_y < 0$ $E_y > 1$ $E_y < 1$ $E_y = 1$	Normal good Inferior good Income elastic Income inelastic Income unit elastic
Price elasticity of supply	$\frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$	$E_s > 1$ $E_s < 1$ $E_s = 1$ $E_s = \infty$ $E_s = 0$	Elastic Inelastic Unit elastic Perfectly elastic Perfectly inelastic

Who Pays the Tax?

A tax placed on the sellers of VCR tapes shifts the supply curve from S1 to S2 and raises the equilibrium price from $8 to $8.50. Part of the tax is paid by buyers through a higher price paid, and part of the tax is paid by sellers through a lower price kept.



Different Elasticities and Who Pays the Tax